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AUTOSAR FiM User Manual

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1 Overview

본 문서는 FiM 사용을 위해 AUTOSAR 플랫폼 사용할 때, 사용자가 파라미터 설정 또는 시스템 설계를 할 때 주의하거나 참고할 사항을 제공한다. Autosar 표준 SRS/SWS 를 기반으로 작성 되었으며, 모듈 사용시 보다 자세한 기능적인 설명이 필요한 경우, 아래 Reference 문서를 참고한다.

설정관련 Category 의 해석은 다음과 같다.

- Changeable (C): User 에 의해서 설정 가능한 항목
- Fixed (F): User 에 의한 변경이 불가한 항목
- NotSupported (N): 사용되지 않는 항목

2 Reference

Sl. No.	Title	Version
1.	AUTOSAR BSW Service API Guide.doc	1.0.0 or later
2.	AUTOSAR_SWS_DiagnosticEventManager.pdf	4.2.0
3.	AUTOSAR_SWS_DevelopmentErrorTracer.pdf	2.2.0
4.	AUTOSAR_SWS_FunctionInhibitionManager.pdf	3.2.0
5.	AUTOSAR_SWS_DiagnosticCommunicationManager.pdf	4.2.0
6.		
7.		

3 Acronyms and abbreviations

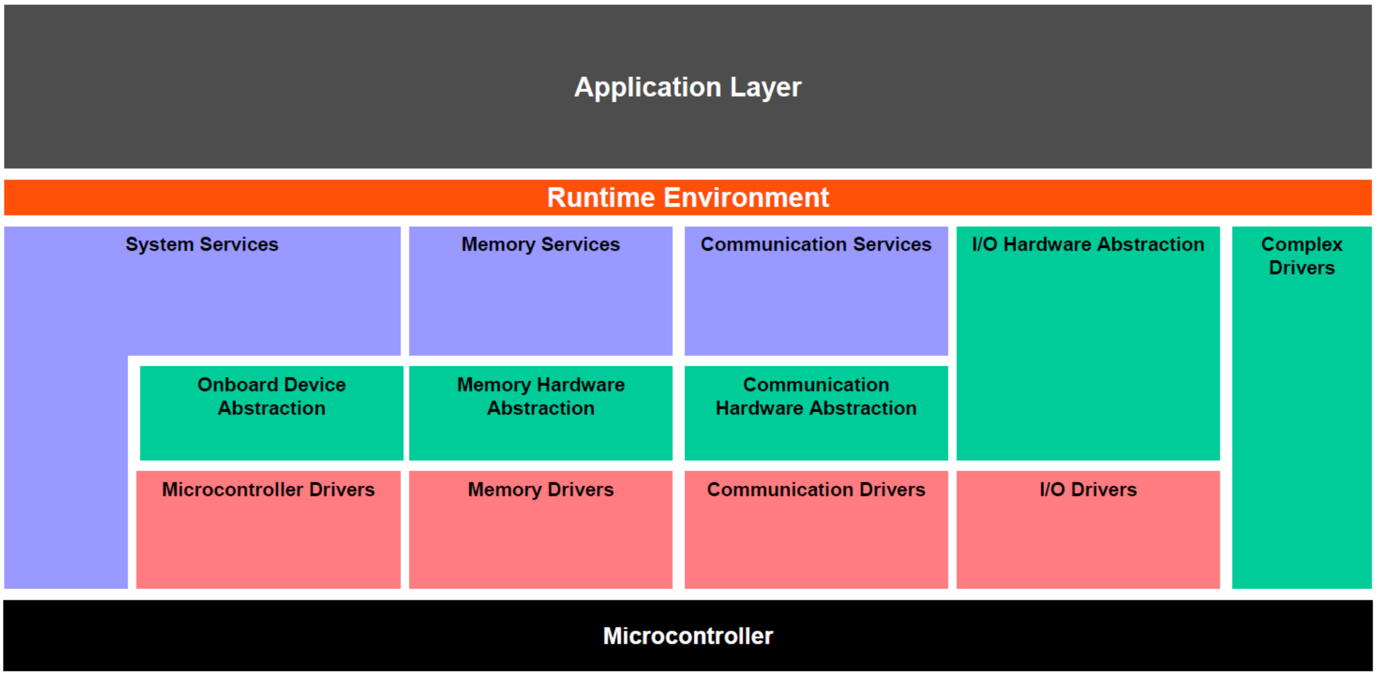
Acronym:	*Description:*
N_OK	Not OK
PossibleErrors	PossibleErrors means the ApplicationErrors as defined in meta model
Application Layer	The Application Layer is placed above the RTE. Within the Application Layer the AUTOSAR Software-Components are placed.

Abbreviation:	*Description:*
API	Application Programming Interface
BSW	Basic Software
CRC	Cyclic Redundancy Check
Dcm	Diagnostic Communication Manager
Dem	Diagnostic Event Manager
Det	Development Error Tracer
DID	Data Identifier
Dlt	Diagnostic Log and Trace
DTC	Diagnostic Trouble Code
ECU	Electronic Control Unit
EcuM	Electronic Control Unit Manager
FDC	Fault Detection Counter
FiM	Function Inhibition Manager
HW	Hardware
ID	Identification/Identifier
ISO	International Standardization Organization
IUMPR	In Use Monitoring Performance Ratio
MIL	Malfunction Indication Light
NVRAM	Non volatile RAM
OBD	Onboard Diagnostics
OEM	Original Equipment Manufacturer (Automotive Manufacturer)
OS	Operating System
PID	Parameter Identification
PTO	Power Take Off
RAM	Random Access Memory
ROM	Read-only Memory
RTE	Runtime Environment
SSCP	synchronous server call point
SW	Software
SW-C	Software Component
UDS	Unified Diagnostic Services

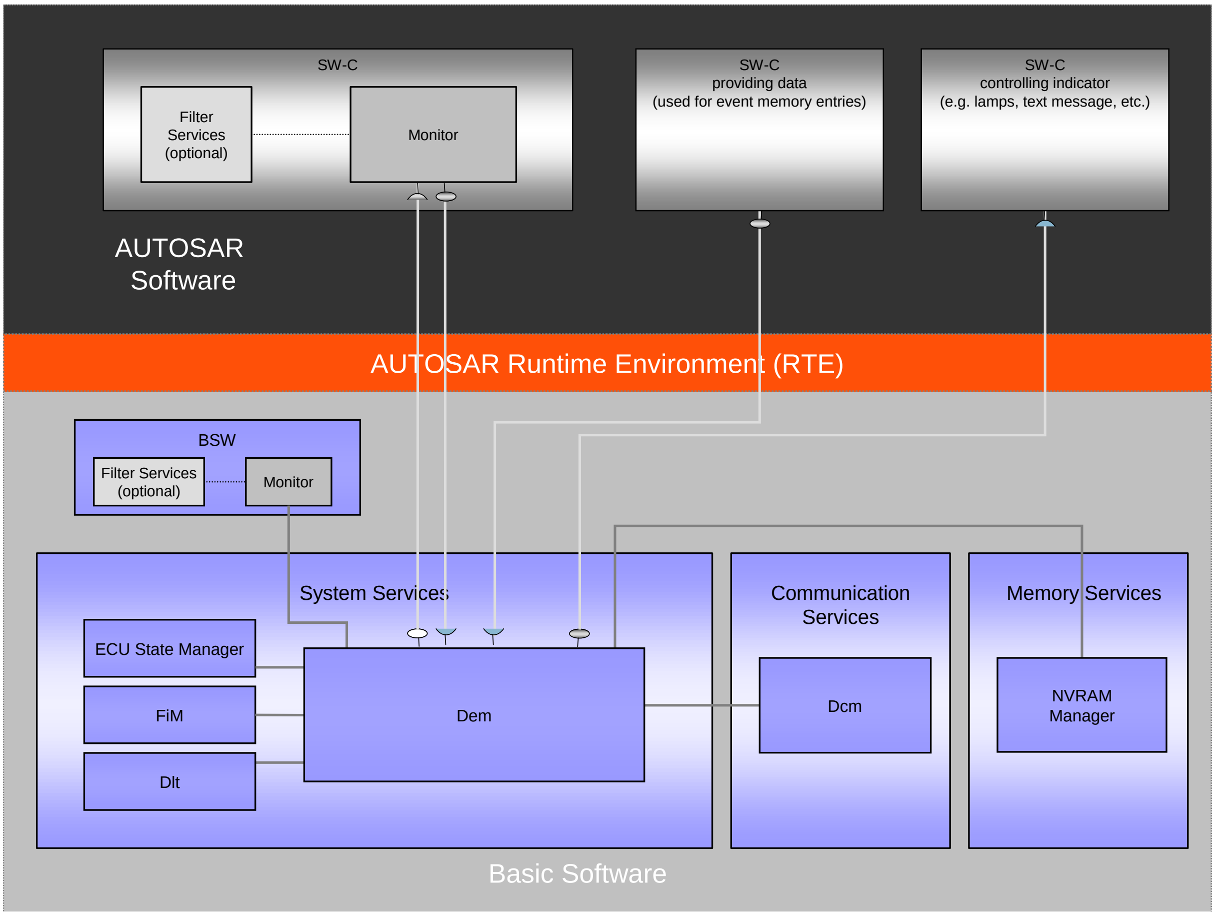
4 AUTOSAR System

4.1 Overview of Software Layers

AUTOSAR 플랫폼의 Layered Architecture 는 아래와 같다. AUTOSAR 플랫폼은, Service Layer, ECU Abstraction Layer, Complex Device Drivers 및 Microcontroller Abstraction Layer 로 구분될 수 있다.



4.2 AUTOSAR Diagnostic Stack



4.2.1 Function Inhibition Manager

Event Status (TestFailed 등) 에 따라 SW-C functionality 의 permission 상태가 변경된다.
SW-C 에서 functionality 의 permission 의 상태를 모니터링하여 functionality 의 동작 유무를 결정한다.

4.2.2 Diagnostic Event Manager

SW-C 및 BSW 모듈에서 발생한 이벤트를 처리한다.

4.2.3 Diagnostic Communication Manager

Diagnostic data flow, diagnostic state 를 관리하며, 진단기의 진단 요청 수행한다.

4.2.4 Development Error Tracer

개발중 발생하는 에러를 관리한다. (양산시 모듈 제거)

5 Product Release Notes

5.1 Overview

이 Chapter 에서는, 현대오트론 Diagnostic Products 에 대한 release 관련 내용을 제공하는데 목적이 있으며, Diagnostic Stack Software product release version 에 대한, 제한사항 및 특이사항을 기술하고 있다.

5.2 Scope of the release

Module	Autosar version	SWS version	Module version
FiM	4.0.3	2.2.0	1.1.2.0

5.2.1 Module release notes

5.2.2 Change Log

- Version 1.1.2.0 (2020-12-30)
  - 개선 사항
    - Misra C 코드 수정

원인	Misra C violation 코드 수정
동작 영향	없음
설정 영향	없음
ASW 조치 사항	없음

- Version 1.1.1.0 (2018-10-11)
  - 개선 사항
    - 코드 공개를 위한 설정 항목 속성 변경

원인	코드 공개에 따라 설정 항목 속성 변경 필요
동작 영향	없음
설정 영향	없음

ASW 조치 사항	없음

➢ Version 1.1.1 (2018-02-28)
- 개선 사항
  ■ SWP Lib 영향성 제거

원인	SWP Lib 영향성 있는 변수 타입 변경
동작 영향	없음
설정 영향	없음
ASW 조치 사항	없음

➢ Version 1.1.0 (2016-09-30)
- 개선 사항
  ■ FiM 초기화 순서 변경(AUTOSAR 4.2.2 선적용)

원인	Dem_Init 에서 FiM_DemTriggerOnEventStatus 호출 시 불필요한 DET Error 발생.(FiM 초기화 시퀀스 변경) AUTOSAR 4.3.0 에 명기된 순서에 따라 초기화 순서를 조정하여 Det 에러가 발생하지 않도록 수정
동작 영향	FiM_Init 과 Dem_Init 초기화 순서 변경(BswM) Dem 2.3.0 이상 필요
설정 영향	없음
ASW 조치 사항	없음

➢ Version 1.0.6 (2016-01-14)
- 개선 사항
  ■ AUTRON_AUTOSAR_Dem_ECU_Configuration_PDF.arxml : Fixed parameter 추가

원인	AUTRON_AUTOSAR_Dem_ECU_Configuration_PDF.arxml : Fixed parameter 추가
동작 영향	없음
설정 영향	없음
ASW 조치 사항	없음

➢ Version 1.0.0 (2015-08-25)
- 개선 사항
  ■ Module version check macro 수정

원인	Module version check macro 수정
동작 영향	없음
설정 영향	없음
ASW 조치 사항	없음

5.2.3 Limitations

- Pre-Compile 만 지원한다.
- **SRS 와 관련된 설정을 수정하고자 할 경우 플랫폼이 재배포 되어야 한다.**
  임의로 수정할 경우 비정상적으로 동작할 수 있다.

5.2.4 Deviation

6 Configuration Guide

(1) 특별한 표기가 없을 경우 AUTOSAR 사양에 근거한 Parameter 이며
HYUNDAI AUTRON 에서 추가한 Parameter 일 경우 (AUTRON specific) 을 표기하였다

(2) Not supported 이면서 default 값이 있는 경우는 설정된 값이 변경되어서는 안된다.

6.1 General

6.2 FiMGeneral

다음 설정을 참고한다.

Parameter Name	Value	Category
FiMMaxEventFidLinks[(1)]		C
FiMMaxEventsPerFid[(2)]		C
FiMMaxFidsPerEvent[(3)]		C
FiMMaxSummaryEvents[(4)]		C
FiMMaxSummaryLinks[(5)]		C
FiMMaxTotalLinks[(6)]		C
FiMDevErrorDetect		C
FiMEventUpdateTriggeredByDem		C
FiMTaskTime		C
FiMVersionInfoApi		C
FiMDataFixed	false	N

(1) FiMMaxEventFidLinks:
This configuration parameter specifies the total maximum number of links between EventIds and FIDs.

(2) FiMMaxEventsPerFid:
This configuration parameter specifies the maximum number of EventIds that can be linked to a single FID.

(3) FiMMaxFidsPerEvent:
This configuration parameter specifies the maximum number of FIDs that can be linked to a single event.
(4) FiMMaxSummaryEvents:
This configuration parameter specifies the maximum number of summarized events that can be configured.

(5) FiMMaxSummaryLinks:
This configuration parameter specifies the total maximum number of links between EventIds and summarized

events.

(6) FiMMaxTotalLinks:
This configuration parameter specifies the total maximum number of links between EventIds and FIDs plus the number of links between EventIds and summarized events.

6.3 FiMConfigSet

다음 설정을 참고한다.

Sub Container(s)	Value	Category
FiMFID(1)		C
FiMEventSummary(2)		C
FiMInhibitionConfiguration(2)		C
FiMSummaryEventId(4)		C

(1) FiMFID:
This container includes symbolic names of all FIDs.

(4) FiMEventSummary:
The summarized EventId definition record consists of a summarized event ID and a specific EventId.
This record means that a particular FID that has to be disabled in case of summarized event (defined above) is to be disabled in any of the specific events. A possible solution could be assigning events as summarized events along with a list of
specific events. During the configuration process the summarized event substitutes the referenced single events.

(3) FiMInhibitionConfiguration:
This container includes all configuration parameters concerning the relationship between event and FID.

(4) FiMSummaryEventId:
This container defines the name of a summarized event.

6.3.1 FiMFID

다음 설정을 참고한다.

Parameter Name	Value	Category
FiMFunctionId(1)		C

(1) FiMFunctionId :
The configuration parameter is used as an ID which represents a functionality. FiMFunctionId is the unique identifier assigned during FIM configuration.

6.3.2 FiMEventSummary

다음 설정을 참고한다.

Parameter Name	Value	Category
FiMInputSumEventRef(1)		C
FiMOutputSumEventRef(2)		C

(1) FiMInputSumEventRef:
Rference to [DemEventParameter].

(2) FiMOutputSumEventRef:

Reference to [FiMSummaryEventId].

6.3.3 FiMInhibitionConfiguration

다음 설정을 참고한다.

Parameter Name	Value	Category
FiMInhInhibitionMask(1)		C
FiMInhFunctionIdRef(2)		C

(1) FiMInhInhibitionMask:
The configuration parameter is used to specify the inhibition mask for [an event – FID] relation.

FIM_LAST_FAILED	Last Failed - DEM_UDS_STATUS_TF flag of Dem Eventstatus is set Use case: Re-configuration, avoiding follow-up errors
FIM_NOT_TESTED	Not Tested this cycle - DEM_UDS_STATUS_TNCTOC flag of Dem Eventstatus is set. Use case: Scheduling of monitors.
FIM_TESTED	Tested - DEM_UDS_STATUS_TNCTOC flag of Dem Eventstatus is not set. Use case: Self deactivation, check during driving cycle.
FIM_TESTED_AND_FAILED	Tested and Failed - DEM_UDS_STATUS_TF flag of Dem Eventstatus is set and DEM_UDS_STATUS_TNCTOC flag is not set Use case: Avoiding deadlocks, repeated monitoring.

(2) FiMInhFunctionIdRef:
Reference to [FiMFID]

Sub Container(s)	Value	Category
FiMInhEventId(1)		C

(1) FiMInhEventId :
The configuration parameter is used for an existing DEM event and summarized events as well.

6.3.3.1 FiMInhEventId

The configuration parameter is used for an existing DEM event and summarized events as well.

다음 설정을 참고한다.

Sub Container(s)	Value	Category
FiMInhRefChoice	User Defined	C

6.3.3.1.1 FiMInhRefChoice

다음 설정을 참고한다.

Sub Container(s)	Value	Category

Sub Container(s)	Value	Category
FiMInhChoiceDemRef		C
FiMInhChoiceSumRef		C

6.3.3.1.1.1 FiMInhChoiceDemRef

다음 설정을 참고한다.

Parameter Name	Value	Category
FiMInhEventRef(1)		C

(1) FiMInhEventRef :
Reference to [DemEventParameter]

6.3.3.1.1.2 FiMInhChoiceSumRef

다음 설정을 참고한다.

Parameter Name	Value	Category
FiMInhSumRef(1)		C

(1) FiMInhSumRef :
Reference to [FiMSummaryEventId]

6.3.4 FiMSummaryEventId

다음 설정을 참고한다.

Parameter Name	Value	Category
FiMEventSumId(1)		C

(1) FiMEventSumId :
The summarized EventId definition record defines the existence of a summarized event with a specific name. This summarized event can be referenced in the FiMEventSummary (as FiMSummaryEventId) and Inhibition configuration (as FiMInhEventId).

6.4 System Configuration

6.4.1 ApplicationSwComponentType 설정

※ AUTOSAR BSW Service API Guide.doc 문서를 참조한다.

6.4.2 CompositionSwComponentType 설정

※ AUTOSAR BSW Service API Guide.doc 문서를 참조한다.

7 Application Programming Interface (API)

7.1 Type Definitions

7.1.1 FiM_FunctionIdType

Type:	uint8/uint16
Range:	0 to 255/0 to 65535
Description:	Type for the FunctionID

7.2 Interfaces

7.2.1 FunctionInhibition

Note: A client can query the FIM for execution permission for a specific function. Functions are represented in the FIM by FunctionIds (FIDs). These FIDs are not directly used by the client SW-C. Instead, the mechanism of "port-defined argument values" is used and every FID is mapped to a separate port that is responsible for the data exchange via RTE

7.2.1.1 GetFunctionPermission

Function Name	Xxx_GetFunctionPermission	
Syntax	Std_ReturnType Xxx_GetFunctionPermission(boolean* Permission)	
Sync/Async	Synchronous	
Reentrancy	Reentrant	
Parameters (In)	None	
Parameters (Inout)	None	
Parameters (Out)	Permission	TRUE: FID has permission to Run FALSE: FID has no permission to Run (shall not be executed).
Return Value	Std_ReturnType	E_OK: The request is accepted E_NOT_OK: The request is not accepted, i.e. initialization of FIM not completed
Description	This service reports the permission state to the functionality.	
Preconditions	None	
Configuration Dependency	None	

7.2.2 참고사항

7.2.2.1 In Communication with application SW-C

RTE 기반 생성된 함수의 프로토타입에 대한 사항은 AUTOSAR BSW Service API Guide.doc 문서 참조.

8 Generator

8.1 Generator Option

Option	Description
-H/-Help	To display help regarding usage of the tool.
-O/-Output	To generate the output files in the specified directory location.
-V/-Version	To display the copyright information and the tool version.
-L/-Log	To generate ₩"$BswConfig::Lis_File_Name₩" file.
-D/-DryRun	To execute in validation mode.
-I/-Info	To disable Information Messages.
-W/-Warn	To disable Warning Messages.
-DDT	To disable the generation of Date and Time Information in the Tool Generated Output Files.

8.2 Generator Error Message

This section helps to analyze the errors or warnings displayed during the execution of the tool. It ensures conformance of input file(s) with syntax and semantics.

The Generation Tool displays errors or warnings or information when the user has configured incorrect inputs. The format of Error/Warning/Information message is as shown below:

- ERR/WRN/INF<mid><xxx>: < Error/Warning/Information Message>
  Where,
  <mid>: 11 – FiM Module Id (11) for user configuration checks.
  000 - for command line checks.
  <xxx>: 001 - 999 - Message ID.
- File Name : Name of the file in which the error has occurred
- Path : Absolute path of the container in which the parameter is present

'File Name' and 'Path' are optional.

Below section provides the list of module specific error, warning and information messages.

8.2.1 Error Messages

ERR011002: Unexpected Error Found. This error may be due to the incorrect configuration of the element(s) <Parameter Name/ Container Name>. If the error is not resolved, then please contact AUTRON AUTOSAR Support System.

This error may occur due to incorrect configuration of the Parameter Name/ Container Name provided in the error message. If the error is not resolved, then contact AUTRON AUTOSAR Support System.

ERR011003: 'Component Name' Component is not present in the input file(s).

This error occurs, if any one of FiM or Dem component is not present in any of the input ECU Configuration Description File(s).

ERR011004: The reference path is empty for the parameter 'Parameter Name' in the container 'Container Name'.

This error occurs, if reference path is not provided for the reference parameter.

Container Name	Parameter Name
FiMEventSummary	FiMInputSumEvenRef
	FiMOutputSumEvenRef
FiMInhibitionConfiguration	FiMInhFunctionIdRef
FiMInhChoiceDemRef	FiMInhEventRef
FiMInhChoiceSumRef	FiMInhSumRef

ERR011005: The parameter 'Parameter Name' in the container 'Container Name' should be configured.

This error occurs, if any of the mandatory configuration parameters mentioned below is not configured in ECU Configuration Description File.

Container Name	Parameter Name
FiMGeneral	FiMDataFixed
	FiMDevErrorDetect
	FiMEventUpdateTiggeredByDem
	FiMMaxEventFidLinks
	FiMMaxEventsPerFid
	FiMMaxFidsPerEvent
	FiMMaxSummaryEvents
	FiMMaxSummaryLinks
	FiMMaxTotalLinks
	FiMTaskTime
	FiMVersionInfoApi
FiMFID	FiMFunctionId
FiMEventSummary	FiMInputSumEvenRef

Container Name	Parameter Name
	FiMOutputSumEvenRef
FiMInhibitionConfiguration	FiMInhInhibitionMask
	FiMInhFunctionIdRef
FiMInhChoiceDemRef	FiMInhEventRef
FiMInhChoiceSumRef	FiMInhSumRef
FiMSummaryEventId	FiMEventSumId
BSW-IMPLEMENTATION	AR-RELEASE-VERSION
	VENDOR-ID
	SW-VERSION
BSW-MODULE-DESCRIPTION	MODULE-ID

ERR011006: The value configured for the parameter 'Parameter Name' in the container 'Container Name' should follow the pattern: <Pattern>
This error occurs, when the parameter 'Parameter Name' is not configured as per the pattern.

Parameter Name	Container Name	Pattern	Example
AR-RELEASE-VERSION	BSW-IMPLEMENTATION	<4.[0-9]+.[0-9]+>	4.0.3
SW-VERSION			

ERR011013: The reference path <Reference Path> provided for the parameter 'Parameter Name' in the container 'Container Name' is incorrect.

This error occurs, if incorrect reference is provided for the reference parameter.

Container Name	Parameter Name
FiMEventSummary	FiMInputSumEvenRef
	FiMOutputSumEvenRef
FiMInhibitionConfiguration	FiMInhFunctionIdRef
FiMInhChoiceDemRef	FiMInhEventRef
FiMInhChoiceSumRef	FiMInhSumRef

ERR011051: Value configured for the parameter 'Parameter Name' in the container 'Container Name' is(are) not unique.

This error occurs when the value configured for the parameter 'Parameter Name' in the container 'Container Name' are not unique.

Container Name	Parameter Name
FiMFID	FiMFunctionId
FiMEventSummary	FiMInputSumEventRef
FiMSummaryEventId	FiMEventSumId

8.2.2 Warning Messages

**WRN011051: The number of DemEventIds referred by the parameter 'FiMInputSumEventRef' in the container 'FiMEventSummary' should be less than the value configured for the parameter 'FiMMaxSummaryLinks' in the FiMGeneral
Container.**

This warning occurs, if the number of DemEventIds referred by the parameter 'FiMInputSumEventRef' is more than value configured for the parameter 'FiMMaxSummaryLinks'

WRN011052: The number of 'FiMInhibitionConfiguration' containers configured is greater than the value configured for parameter 'Parameter Name' in the container 'FiMGeneral'.

This warning occurs, if the number of 'FiMInhibitionConfiguration' containers configured is more than the value configured for the parameter 'Parameter Name' in the container 'FiMGeneral'.

Parameter Name
FiMMaxEventFidLinks
FiMMaxTotalLinks
FiMMaxSummaryEvents

WRN011053: Total number of events configured per FID configured in the container 'FiMInhibitionConfiguration' should be less than or equal to 'FiMMaxEventsPerFid' in the container 'FiMGeneral'.

This warning occurs, if the links between Event ids and Fid in the container 'FiMInhibitionConfiguration' is greater than the value configured for the parameter 'FiMMaxEventsPerFid' in the FiMGeneral container.

WRN011054: The parameter 'FiMEventSumId in the container 'FiMSummaryEventId' is configured but not referred to, in the parameter 'FiMOutputSumEventRef' in the container 'FiMEventSummary'.

This warning occurs, if the parameter 'FiMEventSumId' in the container 'FiM SummaryEventId' is not referred even once.

WRN011055: The parameter 'FiMFunctionId' in the container 'FiMFID' is configured but not referred to in the parameter FiMInhFunctionIdRef in the container FiMInhibitionConfiguration

This warning occurs, if the parameter 'FiMFunctionId' in the container 'FiMFID' is not referred even once.

WRN011056: Value of the pair for parameter 'FiMFunctionId' in the container 'FiMFID' and Dem event parameters configured in the container 'FiMInhibitionConfiguration' should be unique.

This warning occurs when the pair of DemEvents and Fids linked in the container 'FiMInhibitionConfiguration' is repeated.

8.2.3 Information Messages

INF011015: AUTOSAR Release version <value of the element AR-RELEASE-VERSION>) configured for the parameter 'AR-RELEASE-VERSION' in provided MDT file is not correct. AUTOSAR Release version should be one of the following: 4.0.3.

This information occurs, if the value of the element AR-RELEASE-VERSION present in the Bsw Module Description template is configured other than 4.0.3

9 Det Errror

Detected development errors shall be reported to the Det_ReportError() service of the Development Error Tracer (DET) if Det error dection is enabled.

There is only one operation used as service from Development Error Tracer. In C-style, it looks as follows:
Std_ReturnType Xxx_ReportError(uint8 InstanceId, uint8 ApiId, uint8 ErrorId);
Note: ModuleId can be used in "port defined argument value".

9.1 Error classification

Type or error	*Relevance*	*Related error code*	*Value*
This error is reported if any API (FiM_GetFunctionPermission) is invoked by SW-C before FiM module is initialized.	Development	FIM_E_WRONG_PERMISSION_REQ	0x01
This error is reported if any API (FiM_DemTriggerOnEventStatus) is invoked by Dem before FiM module is initialized.	Development	FIM_E_WRONG_TRIGGER_ON_EVENT	0x02
This error is reported if any API (FiM_GetFunctionPermission) is invoked by SW-C with wrong FID.	Development	FIM_E_FID_OUT_OF_RANGE	0x03
This error is reported if any API (FiM_GetVersionInfo, FiM_GetFunctionPermission) is invoked with NULL pointer.	Development	FIM_E_INVALID_POINTER	0x05
This error is reported if any API (FiM_DemTriggerOnEventStatus) is invoked by Dem with wrong Dem_EventStatusExtendedType.	Development	FIM_E_INVALID_EVENTSTATUSEXTENDEDTYPE	0x06
This error is reported if any API (FiM_DemTriggerOnEventStatus) is invoked by Dem with invalid EventId.	Development	FIM_E_INVALID_EVENTID	0xF2
This error is reported if any API (FiM_MainFunction, FiM_DemInit) is invoked before FiM module is initialized.	Development	FIM_E_UNINIT	0xF1

Type or error	*Relevance*	*Related error code*	*Value*
This error is reported if any API (FiM_Init) is invoked by EcuM and initialization is not successful	Development	FIM_E_INIT_NOT_SUCCESS	0xF0

9.2 Service ID

Dcm function name	*Service ID[hex]*
FiM_Init	0x00
FiM_GetFunctionPermission	0x01
FiM_DemTriggerOnEventStatus	0x02
FiM_DemInit	0x03
FiM_GetVersionInfo	0x04
FiM_MainFunction	0x05

10 Appendix

10.1 Initialization sequence of FiM

The initialization of Dem and Fim shall always follow the below order :

step 0) Dem_PreInit

step 1) Non-volatile memory data has to be available

step 2) Fim_Init (setting up internal variables); after FIM_Init, the Fim is not yet ready to be used.

step 3) Dem_Init: do the internal DEM initialization and use Fim_DemInit to finally initialize the FIM」()

Note: From step 3 onwards, the Dem and Fim are finally initialized and ready to be used.